FreeWalker: a smart insole for longitudinal gait analysis

Baitong Wang[1,*], Kuldeep Singh Rajput[1,*], Wing-Kin Tam[1], Anthony K. H. Tung[2] and Zhi Yang[1]

***Abstract*— Gait analysis is an important diagnostic measure to investigate the pattern of walking. Traditional gait analysis is generally carried out in a gait lab, with equipped force and body tracking sensors, which needs a trained medical professional to interpret the results. This procedure is tedious, expensive, and unreliable and makes it difficult to track the progress across multiple visits. In this paper, we present a smart insole called FreeWalker, which provides quantitative gait analysis outside the confinement of traditional lab, at low- cost. The insole consists of eight pressure sensors and two motion tracking sensors, i.e. 3-axis accelerometer and 3-axis gyroscope. This enables measurement of under-foot pressure distribution and motion sequences in real-time. The insole is enabled with on-board SD card as well as wireless data transmission, which help in continuous gait-cycle analysis. The data is then sent to a gateway, for analysis and interpretation of data, using a user interface where gait features are graphically displayed. We also present validation result of a subject's left foot, who was asked to perform a specific task. Experiment results show that we could achieve a data-sampling rate of over 1 KHz, transmitting data up to a distance of 20 meter and maintain a battery life of around 24 hours. Taking advantage of these features, FreeWalker can be used in various applications, like medical diagnosis, rehabilitation, sports and entertainment.**

I. INTRODUCTION

Gait analysis is the systematic and quantitative evaluation of parameters that characterize human locomotion [1]. It has been widely used in various applications such as medical diagnostics [2], rehabilitation [3] and sports [4]. For instance, gait analysis is an important tool to detect fall risk and fall prevention in ageing population [5]. In orthopedic and rehabilitation, gait analysis is used to keep a track of patient after surgery and also useful in application, which involve orthopedic assistive devices.

The standard method to analyze gait is based on the multi- camera motion capture system and a platform, which is capable of measuring ground-reaction forces. When the subject/patient walks along the platform, his motion patterns are captured by the camera and the sensors embedded onto the floor. However, this type of motion capture system is usually restricted to indoor use as motion capture camera often rely on infrared reflective markers that will be affected by sunlight. The high cost of the equipment also makes gait analysis prohibitive for wide adoption. Moreover, using the data captured by these devices, the trained expert needs to analyze the walking pattern manually. So this traditional method of gait analysis is expensive and time-consuming, as it required post-processing time. Upon all, the patient feels uncomfortable during the study and hence the sensing data might not be accurate. Hence there is a need for solutions, which can be used by patients, especially elderly, to remotely monitor their gait in a more intuitive and reliable manner.

With the development of wearable sensors, gait analysis becomes quite inexpensive and the monitoring can be performed in comfortable and realistic settings, outside the lab. There has been prior research on using sensors like accelerometers and gyroscopes to analyze various gait features [6]. Chen et al. [7] proposed a longitudinal motion assessment method using inertial sensors on cerebral palsy patients and Bae et al. [8] designed a Force Sensing Resistive (FSR) sensor array based gait analysis. Researchers have been working on designing a pair of instrumented insoles with piezoresistive pressure sensor and three-axis accelerometers, which can measure, pressure distribution of the foot during various gait phases [9]. Recently, Xu et al. [10] proposed an insole, which consists an array of 48 pressure and motion sensing sensors to analyze gait parameters. Although these methods help us understand the walking pattern efficiently, they cannot address the problem of analyzing gait cycle and its features accurately in real-time.

In this paper, we present an insole called FreeWalker, which addressed the following issues faced in current gait monitoring systems i.e. sampling rate, power consumption and real-time gait analysis. The insole consists of eight disc shaped pressure sensors along with motion sensing components embedded into the insole. With wireless data transfer at the rate of 1 KHz, we could accurately analyze various gait features and detect gait cycle in real-time, maintaining a battery backup of more than 24 hours. We also present a Graphical User Interface, which shows foot pressure distribution maps, and various gait parameters graphically. This framework allows the patient to continuously analyze their locomotion remotely, at low cost.

The remainder of this paper is organized as follows; Section II descries the overall framework of the smart insole, covering hardware and software architecture. In Section III, we present experimental results with data analysis and finally conclude with future directions and possible applications in Section IV.

II. SYSTEM OVERVIEW

A. Framework

Identifying normal and pathological walking patterns is a complicated process and requires analysis of various gait features like cadence, stride height, stride length and speed, which are very basic features. But if we want to use

*These authors have equal contribution to this paper.

[1]Department of Electrical and Computer Engineering, National University of Singapore, [2] Department of Computer Science, National University of Singapore correspondence: Zhi Yang, eleyangz@gmail.com

978-1-4244-9270-1/15/$31.00 ©2015 IEEE

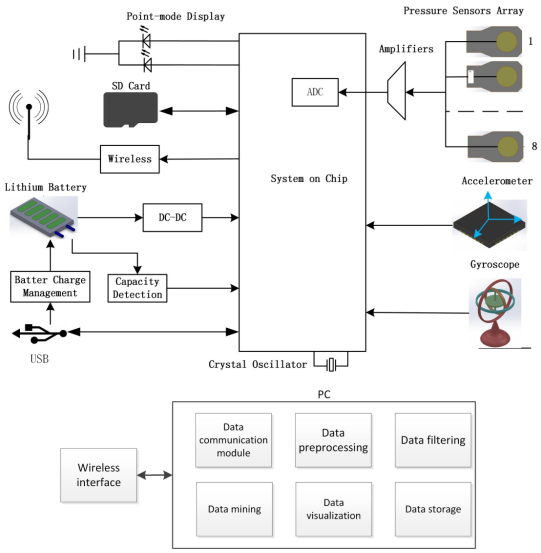

Fig. 1. Overview of insole system architecture.

this methodology for foot and ulcer prevention in diabetes patients, or analyze fall prevention, we need to measure the pressure distribution and pressure balance between two feet. Hence, different gait features are needed for different kinds of application.

In context to medical application, each gait cycle is divided into four stages: swing, contact, midstance and propulsive [11]. Our main goal is to design an insole, which is capable of identifying the gate stage in real-time and detect various gait parameters, required to analyze human locomotion. To address this problem, we came up with an overall system design which is divided into three major parts, the insole hardware design, data acquisition and transmission and finally the data processing and graphical user interface as shown in Fig. 1.

B. Hardware and software architecture

The insole consists of 8 low-cost pressure sensors of diameter 5mm, embedded into it, to detect and measure a relative change in the force or applied load as shown in Fig. 2(a). As shown in Fig. 2(b), these sensors are positioned in a unique manner, such that the spatial pressure distribution of the whole foot is extracted. It also consists of motion tracking sensors like the 3-axis accelerometer and gyroscope. All these sensors are connected to an ARM based microcontroller. The pressure sensor acts as a force-sensing resistor in an electric circuit. We integrate these sensors into our system design by incorporating it into a force-to-voltage circuit, Fig. 2(c). The circuit uses an inverting operating amplifier arrangement to produce an analog output based on the sensor resistance and a fixed reference resistance (R2). We then connect the output to analog-to-digital converter to change this voltage to a digital output as shown in Fig. 1. In the circuit, the sensitivity of the sensor could be adjusted by changing the reference resistance and/or drive voltage. The circuit is adjusted to operate at full force range (0-100 lb.) and response time of less than 5μsec.

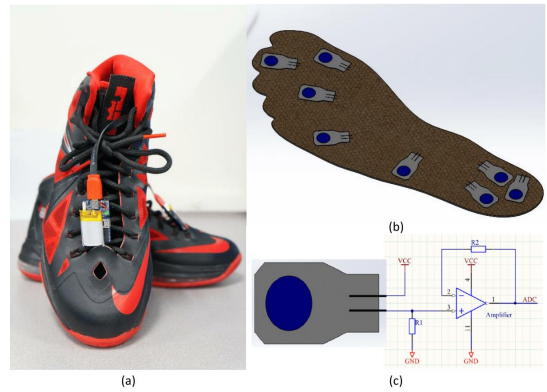

Fig. 2. Insole design (a) Insole Prototype with the embedded hardware (b) Sensor position (c) Force to voltage circuit (sensor circuit).

We calibrate the pressure sensor to convert the output voltage into the appropriate engineering unit. We use a pressure-calibration device to apply a known force o the sensor and equate the sensor circuit output voltage to this force. We repeat this step and finally plot a pressure versus collage curve, which is linear as shown in Fig. 3.

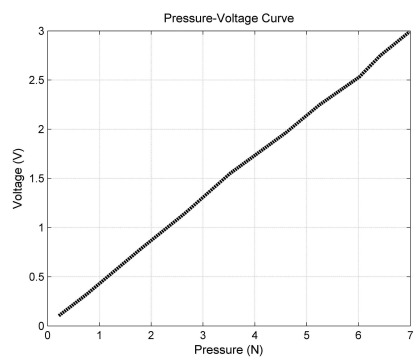

Fig. 3. Pressure sensor calibration curve.

To get more insights into the gait parameters we use a motion-tracking device (MPU-6050 by InvenSense, USA), which combines 3-axis accelerometer and 3-axis gyroscope in a small 4x4x0.9mm package. A sensor fusion algorithm has been implemented in the Digital Motion Processor on the MPU-6050 to obtain the orientation of our device based on accelerometer and gyroscope signals. This tiny compact chip features three 16-bit analog-to digital converters for digitizing the gyroscope and accelerometer output. The sensors selected for the insole are detailed in Table 1.

The second part of the design is the data acquisition and transmission. The system has an on-board SD card storage where all the sensor data is continuously stored. We use wireless transmission to send the data in real-time to a dedicated gateway (person computer) for further data analysis. We design an integrated wireless system as shown in Fig. 4, using the wireless chip (nRF24L01 by Nordic Semiconductor, Norway) which is highly integrated, ultra-low-power and the data transfer rate is faster than Zigbee and Bluetooth, which meets our requirement of medical grade

TABLE I

SENSORS SELECTED FOR FREEWALKER

Sensor Type	Parameters	Sensor output	Operating range
Pressure sensor (FlexiForce, #A301)	Pressure distribution, heal-strike and toe-off timing, gait cycle analysis	Act like a force sensing resistor. Use force to voltage circuit.	Medium force rage (110N= 11.22Kg)
Accelerometer (InvenSence, MPU-6050)	Stride height and length, Cadence	Voltage change corresponding to acceleration (velocity and distance can be calculated)	Full scale rage of ±2g
Gyroscope (InvenSence, MPU-6050)	Foot orientation (roll, pitch, yaw)	Voltage proportional to angular velocity (Angle of rotation can be calculated)	Full scale range, $\pm 2000^o$/sec

gait analysis. We achieve data sampling frequency of more than 1 KHz for the pressure sensors and around 100Hz for the accelerometer and gyroscope, whose sampling frequency is limited by MPU-6050. We also have an efficient power management system to improve the battery life (Li-battery), i.e. the power switching circuit turns on the power only when the command for data transmission is encountered. Using this system design we could achieve data transmission distance of more than 20 meters maintaining the battery backup of over 24 hours.

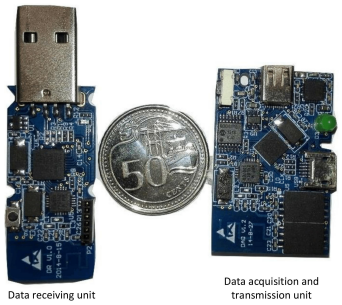

Fig. 4. Custom designed acquisition, transmission and receiving unit.

The third key component of our system architecture is the data analysis and display. We develop a software interface (GUI) using LabView (National Instruments, USA), to analyze and display the raw sensor data it receives. The wireless receiver as shown in Fig. 4, uses the Virtual Serial Port based on the USB 2.0 protocol to feed the data into LabView. The received data is hexadecimal, which is then converted into pressure, acceleration and Euler angles for analysis. In the data mining section, we use multithreading for real-time display of the data in the form of waveforms and pressure maps as shown in Fig. 5. For the analysis, as mentioned previously the software calibrates the raw sensor data, removes the noise and computes various gait features (foot orientation (roll, pitch yaw), cadence, swing, under-foot pressure).

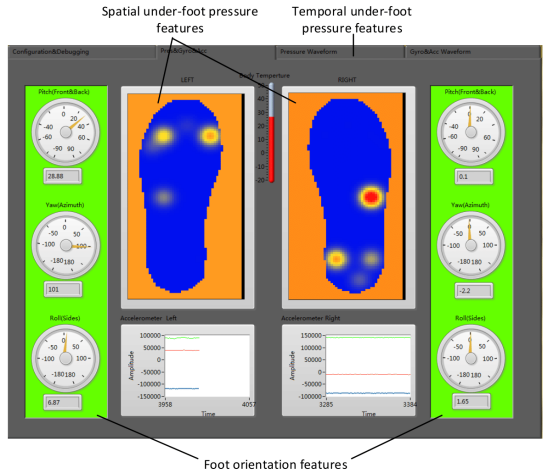

Fig. 5. Graphical user interface design.

III. EVALUATION

We recruited volunteers for the validation of the FreeWalker. The subject had to walk 10 steps forward, take a 'U' turn and walk 10 steps backwards to his original start position, at normal walking speed, which was set as the testing protocol. This experiment helped us evaluate the pressure distribution, acceleration and motion sequences of the subject.

Fig. 6 shows the variation in pressure distribution in the contact, midstance and propulsive stages and it can clearly been seen that, specific pressure sensors are activated in different gait stages. For instance, there is pressure at the lower part of the feet when the heal strikes the ground, i.e. contact stage. We use the pressure sensor data from the front, middle and the back portion of the foot to distinguish between various gait stages in real-time. This is implemented using time-domain classification technique, where each gait cycle is analyzed considering when the users heal strikes the ground, when it is contact and finally when the toe is off the ground, as shown in Fig. 6 and Fig. 7.

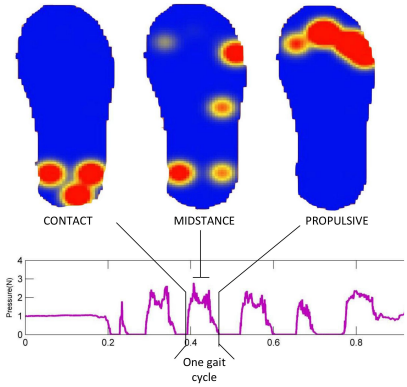

Fig. 6. Pressure map for three stages in one gait cycle.

For simplicity of presentation, we analyze one volunteer's locomotion and visualize the data from his left foot. The analysis was done looking at the 8 pressure sensor data, 3-

axis accelerometer and 3-axis gyroscope. When the subject walks 10 steps forward, his left foot completes 5 gait cycles. Similarly, when he takes a 'U' turn and walks backward, his left foot again completes 5 gait cycles. This can clearly be seen from the Fig. 7, where we plot the pressure distribution curve for various gait cycles.

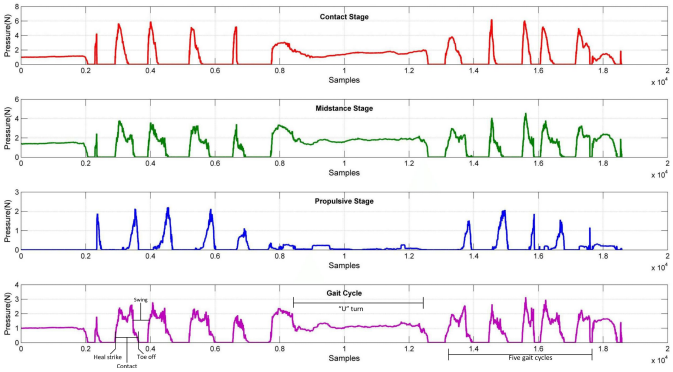

Fig. 7. Pressure distribution data from the subjects left foot while walking at normal speed.

These pressure curves can be used to calculate various gait features like, swing time (transition time between of foot from heal strike to toe off), number of steps and landing position (position of the foot when it lands). We also plot the data from the 3-axis accelerometer and the 3-axis gyroscope as shown in Fig. 8. Visual analysis of these plots helps in interpretation of number of steps the person has walked, exact time and position when he took a 'U' turn (by looking at the change in Yaw), cadence, foot orientation, walking speed, and stride height and length, which help in Gait analysis.

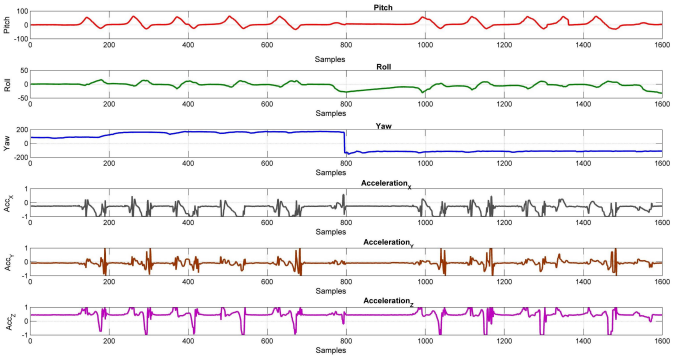

Fig. 8. Accelerometer and Gyroscope data from the left insole for the subject walking at normal speed.

IV. CONCLUSION AND FUTURE WORK

In this paper, we present a smart insole called FreeWalker for ambulatory gait analysis. The system design integrates various technologies like sensor, computing and wireless. We presented preliminary data from a subject using the insole and show that the data can be used to analyze various medical grade gait parameters. These gait analysis results can be transmitted to a centralized medical location and processed by trained medical personnel remotely. Unlike [13], we just use 8 pressure sensors to carry real-time gait analysis, achieving high data sampling rate, large transmission distance and long battery life, making it unique for various application, likes sports, entertainment and rehabilitation.

Our future work would involve the development of further analytics and evaluation of the insole through clinical trials. First, we would come up with algorithms to prevent fall or reduce the risk of fall and provide alert in case of emergency. Second, we plan to develop a system that can automatically detect abnormal gait patterns leading to falls and instability on elderly. We also plan to use this device as a continuous progress-tracking device for rehabilitation. In terms of clinical applications, the insole can be used in hospital wards to detect the motion of the patient enabling the clinician/nurse to track the motion of the patient remotely.

ACKNOWLEDGEMENT

This work is supported by Singapore National Research Foundation under its International Research Center at Singapore Funding Initiative.

REFERENCES

[1] Salim Ghoussayni, Christopher Stevens, Sally Durham, and David Ewins. Assessment and validation of a simple automated method for the detection of gait events and intervals. *Gait & Posture*, 20(3):266–272, 2004.

[2] Michael W Whittle. *Gait analysis: an introduction*. Butterworth-Heinemann, 2014.

[3] Martha E Eastlack, Julianne Arvidson, Lynn Snyder-Mackler, Jerome V Danoff, and Charles L McGarvey. Interrater reliability of videotaped observational gait-analysis assessments. *Physical Therapy*, 71(6):465–472, 1991.

[4] H Stolze, JP Kuhtz-Buschbeck, C Mondwurf, A Boczek-Funcke, K Jöhnk, G Deuschl, and M Illert. Gait analysis during treadmill and overground locomotion in children and adults. *Electroencephalography and Clinical Neurophysiology/Electromyography and Motor Control*, 105(6):490–497, 1997.

[5] Federico Bianchi, Stephen J Redmond, Michael R Narayanan, Sergio Cerutti, and Nigel H Lovell. Barometric pressure and triaxial accelerometry-based falls event detection. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 18(6):619–627, 2010.

[6] Tao Liu, Yoshio Inoue, and Kyoko Shibata. Development of a wearable sensor system for quantitative gait analysis. *Measurement*, 42(7):978–988, 2009.

[7] Shanshan Chen, Christopher L Cunningham, Bradford C Bennett, and John Lach. Enabling longitudinal assessment of ankle-foot orthosis efficacy for children with cerebral palsy. In *Proceedings of the 2nd Conference on Wireless Health*, page 4. ACM, 2011.

[8] Joonbum Bae, Kyoungchul Kong, Nancy Byl, and Masayoshi Tomizuka. A mobile gait monitoring system for gait analysis. In *Rehabilitation Robotics, 2009. ICORR 2009. IEEE International Conference on*, pages 73–79. IEEE, 2009.

[9] Fernando Martínez-Martí, María Sofía Martínez-García, Santiago G García-Díaz, Javier García-Jiménez, Alberto J Palma, and Miguel A Carvajal. Embedded sensor insole for wireless measurement of gait parameters. *Australasian Physical & Engineering Sciences in Medicine*, 37(1):25–35, 2014.

[10] Wenyao Xu, Ming-Chun Huang, Navid Amini, Jason J Liu, Lei He, and Majid Sarrafzadeh. Smart insole: a wearable system for gait analysis. In *Proceedings of the 5th International Conference on PErvasive Technologies Related to Assistive Environments*, page 18. ACM, 2012.

[11] Sheila A Dugan and Krishna P Bhat. Biomechanics and analysis of running gait. *Physical medicine and rehabilitation clinics of North America*, 16(3):603–621, 2005.